

CNN (Convolutional Neural Network)

It is composed of Multiple layer such as, Convolutional layer, Batch Normalization, Pooling, fully connected layer.

① No. of trainable parameters in Convolutional layer = $(K_w \times K_h \times i_{in} + 1) \times i_{out}$

$K_w \rightarrow$ filter width

$K_h \rightarrow$ filter height

$i_{in} \rightarrow$ No. of I/P channels

$i_{out} \rightarrow$ No. of filters (O/P channels)

" $+1$ " $\rightarrow$ for bias term

② No. of trainable parameters in Batch Normalization = $2 \times \text{No. of features \{ obtained in previous layer \}}$

③ Pooling layer (like max, min, Avg. pooling) ~~don't~~ have learnable parameters
So, they contribute '0' to the parameter count

④ Fully connected layer has = $(\text{I/P units} \times \text{output units}) + \text{output units}$

⑤ Combining All layers: To find the total no. of parameters in CNN sum all the parameters from all the layers calculated above.

Example: Consider a CNN with the following layers:

1. Conv layer : 16 filters, 3×3 , 3 I/P channels
2. Conv layer : 32 filter, 3×3 , 16 I/P channels
3. Fully Connected layer : 128 I/P units, 64 O/P units
4. Batch Normalization after each Conv-Layer.

∴ Solution:

Conv Layer 1: $(3 \times 3 \times 3 + 1) * 16 = 448$

Batch Norm 1: $2 * 448 = 896$ $2 * 16 = 32$

Conv Layer 2: $(3 \times 3 \times 16 + 1) * 32 = 4640$

Batch Norm 2: $2 * 32 = 64$

Fully Connected: ~~448~~ $(128 \times 64) + 64 = 8256$

∴ Total Parameters = 13440

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